Answer: $g_{\max} = 998\,285$.

0.0.1 1. Preliminary observations

Let

$$P = a_1 a_2 \dots a_k, \qquad g = \gcd\Big(\frac{P}{a_1}+1, \dots, \frac{P}{a_k}+1\Big).$$

For each i,

$$g \mid \frac{P}{a_i}+1 \quad \Longrightarrow \quad \frac{P}{a_i} \equiv -1 \pmod g.$$

Multiplying by a_i gives

$$P + a_i \equiv 0 \pmod g \qquad (1)$$

hence $g \mid P + a_i$ for every i. Subtracting two such relations, $g \mid a_i - a_j$ for all i, j; in particular all the numbers a_i are congruent modulo g. Write this common residue as

$$a_i = g\,n_i + r \qquad (0 \le r < g)$$

with integers n_i that are distinct (the a_i are distinct). Because $g \mid P + r$ by (1), reducing the product $P \equiv r^k \pmod g$ and using (1) we obtain

$$r^k + r \equiv 0 \pmod g \quad \Longrightarrow \quad g \mid r\big(r^{k-1}+1\big).$$

If $r = 0$ then each a_i is a multiple of g; but then $\frac{P}{a_i}+1 \equiv 1 \pmod g$, contradicting $g > 1$. Hence $r \neq 0$ and r is coprime to g; consequently from (3)

$$\boxed{g \mid r^{\,k-1}+1}.$$

0.0.2 2. The number of terms

From (2) we have

$$\sum_{i=1}^{k} a_i = g \sum_{i=1}^{k} n_i + kr.$$

Since the n_i are distinct non-negative integers, $\sum n_i \ge 0 + 1 + \cdots + (k-1) = \frac{k(k-1)}{2}$; therefore

$$\sum a_i \ge g\frac{k(k-1)}{2} + kr < 3 \cdot 10^6.$$

From (5),

$$g \le \frac{3\cdot 10^6 - kr}{k(k-1)/2} < \frac{3\cdot 10^6}{k(k-1)/2} = \frac{6\cdot 10^6}{k(k-1)}.$$

The right–hand side decreases as k grows, so the largest possible g can occur only for the smallest admissible value $k=3$. For $k=3$ inequality (5) becomes

$$3(g+r) < 3\cdot 10^6 \quad\Longrightarrow\quad \boxed{g+r \le 999\,999}.$$

0.0.3 3. The case $k=3$

With $k=3$, condition (4) gives

$$\boxed{g \mid r^2+1}.$$

Thus $r^2+1 = g\,t$ for an integer $t \ge 1$; t is a divisor of r^2+1 and by (6)

$$\frac{r^2+1}{t} + r = g + r \le 999\,999.$$

For a fixed divisor t the left side of (8) is increasing in r; hence, for that t, the largest admissible r is the greatest integer satisfying

$$\frac{r^2+1}{t} + r \le 999\,999 \qquad\Longleftrightarrow\qquad r^2 + tr + 1 \le t\cdot 999\,999.$$

Solving (9) for r gives the estimate

$$r \le \sqrt{t\cdot 999\,999} \;-\; \frac{t}{2}.$$

0.0.4 4. Maximising g

From (7) we have $g = (r^2+1)/t$. Using (10),

$$g \;=\; \frac{r^2+1}{t} \;\le\; \frac{\left(\sqrt{t\cdot 999\,999} - \frac{t}{2}\right)^2 + 1}{t} = 999\,999 - \sqrt{t\cdot 999\,999} + \frac{t}{2} + \frac{1}{t}.$$

The right–hand side is a decreasing function of t for $t \ge 2$ (its derivative in t is negative). Consequently the greatest possible g is obtained for the smallest admissible t.

- **$t=1$.** Then $g = r^2+1$ and (8) becomes $r^2+r+1 \le 999\,999$; the largest integer r satisfying this is $r = 999$. Hence $g \le 999^2+1 = 998\,002$.
- **$t=2$.** Now r must be odd (otherwise r^2+1 would be odd and could not contain the factor 2). From (9) we obtain $(r+1)^2 \le 2\cdot 999\,999$, i.e.

$$r \le \left\lfloor \sqrt{2 \cdot 999\,999} \; - \; 1 \right\rfloor = 1413.$$

Taking the maximal odd $r = 1413$ gives

$$g = \frac{1413^2 + 1}{2} = 998\,285.$$

- $t \ge 3$. Using (11) with $t = 3$ gives

$$g < 999\,999 - \sqrt{3 \cdot 999\,999} + \frac{3}{2} \; < \; 998\,268.$$

Since the right-hand side is already smaller than the value obtained for $t = 2$, any larger t yields an even smaller upper bound. Thus no $t \ge 3$ can produce a g larger than $998\,285$.

Consequently the absolute maximum of g is $998\,285$, attained when $t = 2$ and $r = 1413$.

0.0.5 5. A concrete construction attaining $g = 998\,285$

Let

$$g = 998\,285, \qquad r = 1413,$$
$$a_1 = r = 1413, \qquad a_2 = g + r = 999\,698, \qquad a_3 = 2g + r = 1\,997\,983.$$

All three numbers are positive and distinct, and

$$a_1 + a_2 + a_3 = 3(g + r) = 2\,999\,094 < 3 \cdot 10^6.$$

Because all $a_i \equiv r \pmod{g}$ and $r^2 + 1 = 2g$, we have $r^2 \equiv -1 \pmod{g}$. Hence for each i,

$$\frac{P}{a_i} \equiv r^2 \equiv -1 \pmod{g} \quad \Longrightarrow \quad g \mid \frac{P}{a_i} + 1.$$

Writing the three quotients explicitly (using $P = a_1 a_2 a_3$)

$$\begin{aligned} \frac{P}{a_1} + 1 &= g\big(2g + 3r + 1\big), \\ \frac{P}{a_2} + 1 &= g\big(2r + 1\big), \\ \frac{P}{a_3} + 1 &= g\big(r + 1\big). \end{aligned}$$

Since

$$\gcd(r + 1, 2r + 1) = \gcd(r + 1, \, 2r + 1 - 2(r + 1)) = \gcd(r + 1, -1) = 1,$$

the three numbers inside the parentheses are pairwise coprime; therefore

$$\gcd\Big(\frac{P}{a_1}+1,\frac{P}{a_2}+1,\frac{P}{a_3}+1\Big)=g.$$

Thus the constructed set indeed yields the greatest common divisor $g = 998\,285$.

0.0.6 6. Conclusion

For any admissible collection of distinct positive integers $a_1,\dots,a_k$ ($k \ge 3$) with sum $< 3\cdot 10^6$, the integer $g=\gcd\big((P/a_1)+1,\dots,(P/a_k)+1\big)$ satisfies $g \le 998\,285$, and the bound is attained (for example by the three numbers 1413, 999698, 1997983).

Hence the maximal possible value of g is

$$\boxed{998\,285}.$$
